EXCITATION TRANSPORT IN ULTRA-COLD RYDBERG GASES

Spectral Signatures of Dipole Blockade

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WHAT'S THE EFFECT OF DIPOLE BLOCKADE ON TRANSPORT?

A SIMPLE BLOCKADED RYDBERG GAS MODEL

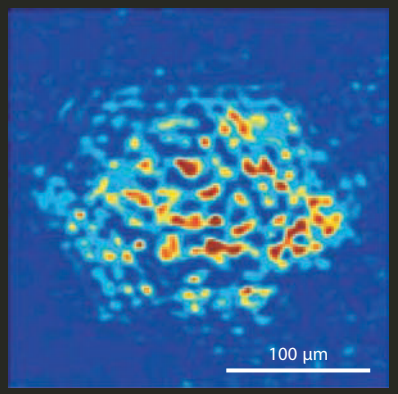
PAIR LOCALIZATION

WHAT'S THE EFFECT OF DIPOLE BLOCKADE ON TRANSPORT?

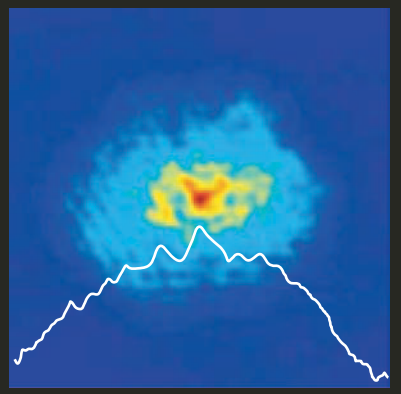
- Rydberg atoms are randomly distributed and interact via strong long-range forces $V \sim R^{-3}$

$\Rightarrow$ dynamics are fundamentally different from short-range interaction in ordered ensembles

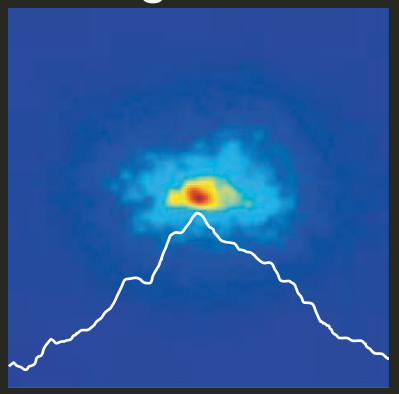
no disorder



some disorder



strong disorder

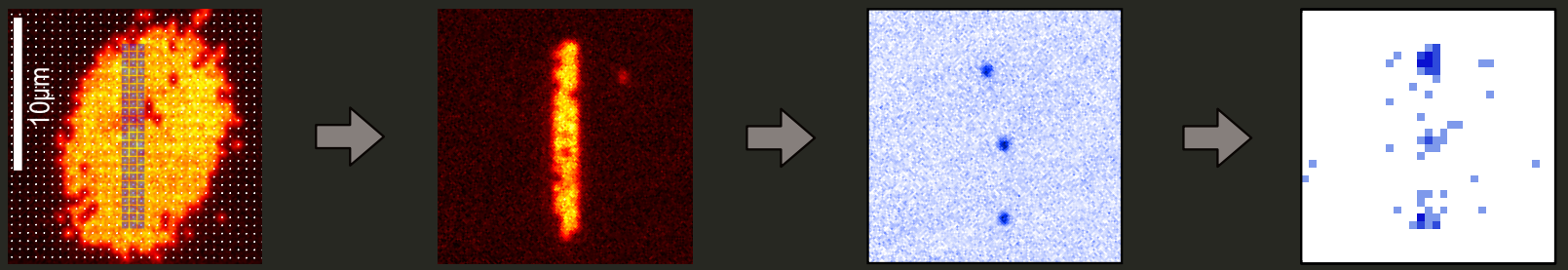


Segev et al., Nat. Photonics **7**, 197–204 (2013)

WHAT'S THE EFFECT OF DIPOLE BLOCKADE ON TRANSPORT?

- dipole blockade effect can create ordered structures from unordered ensembles

⇒ dipole blockade should affect spectral, eigenstate, and transport statistics strongly



Schauß et al., arXiv:1404.0980v2

WHAT'S THE EFFECT OF DIPOLE BLOCKADE ON TRANSPORT?

A SIMPLE BLOCKADED RYDBERG GAS MODEL

PAIR LOCALIZATION

A SIMPLE BLOCKADED RYDBERG GAS MODEL

Rydberg-Rydberg Interactions

- spherical frozen gas cloud with N Rydberg atoms and typical nearest neighbor distance

$$(2\pi\rho)^{-1/3} \simeq 10\mu\mathrm{m}$$

- two Rydberg states per atom:

$$\mathrm{P} = n\mathrm{P}_{3/2,m_j=|3/2|},$$
$$\mathrm{S} = n\mathrm{S}_{1/2,m_j=|1/2|}$$

  with $E_\mathrm{P} - E_\mathrm{S}$ the same for all atoms

- interaction matrix elements:

$$H_{ij} = \langle i|H|j\rangle = \left(1-\delta_{ij}\right)\beta\,\frac{9\sqrt{3}}{8\pi}\left(3\cos^2\Theta_{ij}-1\right)R_{ij}^{-3},$$

  transfers P from atom i to atom j

- choose units of length and energy such that $\rho \equiv \beta \equiv 1$

A SIMPLE BLOCKADED RYDBERG GAS MODEL

Dipole Blockade And Matrix Element Statistics

- H_{ij} is a stochastic variable with a probability density

$$f_{H_{ij}}(h) = \overline{\delta\left(h - H_{ij}\right)}$$

- Rydberg atoms are not point-sized
- Rydberg blockade radius r_b defines an exclusion sphere:

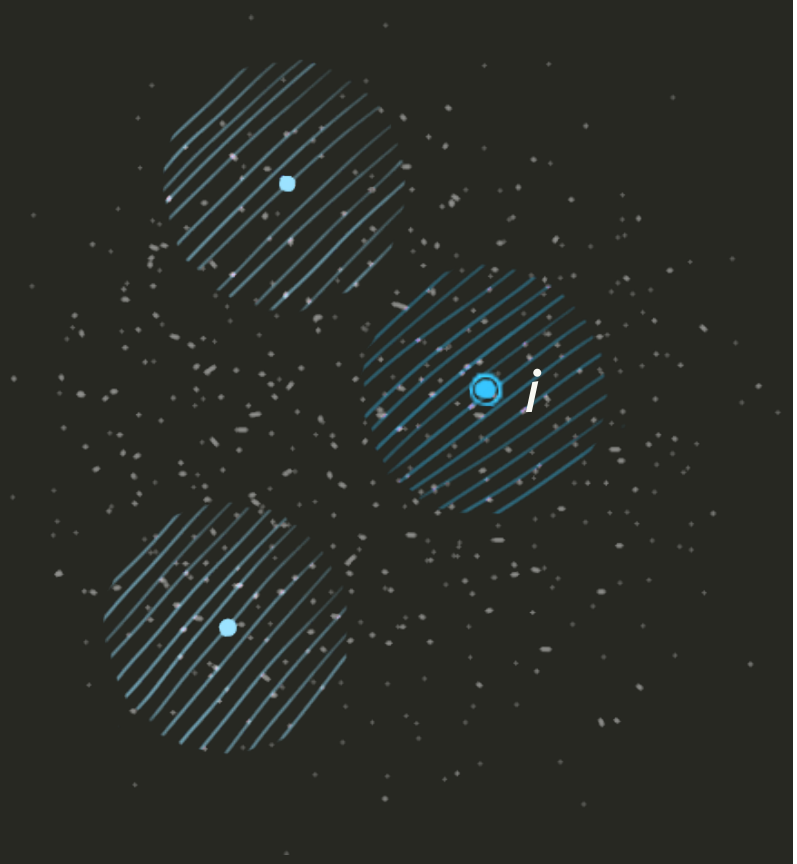
$$r_\mathrm{b} \leq R_{ij} \leq d = 2\left({}^{3N}/_{4\pi}\right)^{1/3}$$

- for $r_\mathrm{b} = 0$: $f_{H_{ij}}$ is heavy-tailed,

$$f_{H_{ij}}(h) \sim_{|h|\to\infty} N^{-1}|h|^{-2}$$

- for $r_\mathrm{b} > 0$: $f_{H_{ij}}$ is truncated,

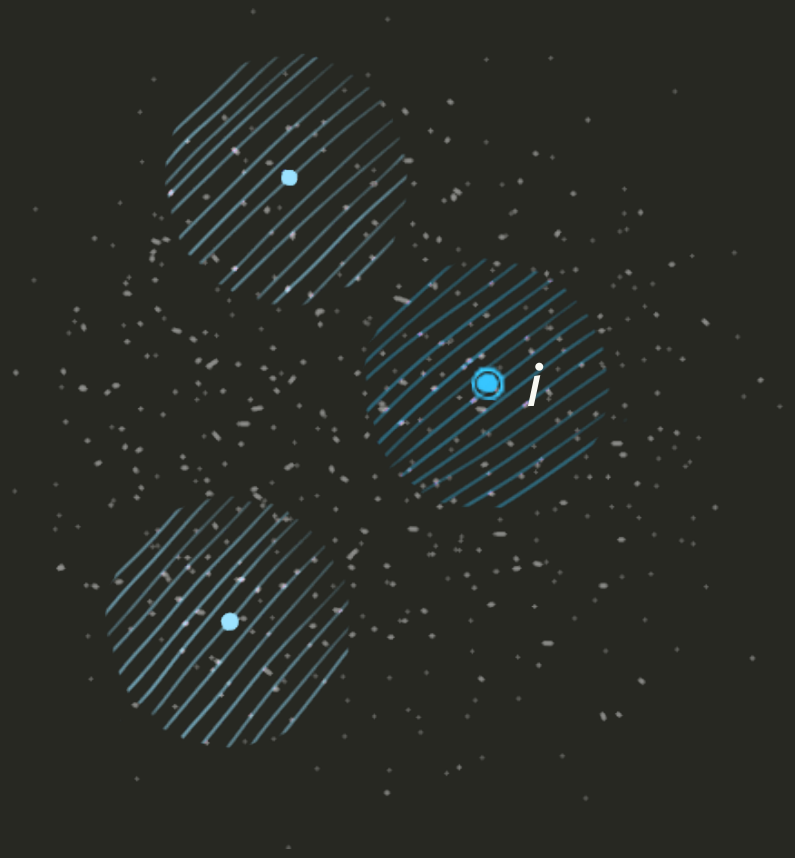
$$\left|H_{ij}\right| \leq {}^{9\sqrt{3}}/_{4\pi\, r_\mathrm{b}^3}$$



A SIMPLE BLOCKADED RYDBERG GAS MODEL

The Localization Problem

- at $t = 0$: P excitation at atom i, all $(N-1)$ other atoms in S
- question: will the P excitation
  - spread to neighboring atoms?
  - spread over the whole cloud?
  - remain localized around atom i?
- here: analyze the spectral statistics of the ensemble
  - easily accessible
  - carries information on the spatial extent of eigenfunctions
- later: full eigenstate statistics



WHAT'S THE EFFECT OF DIPOLE BLOCKADE ON TRANSPORT?

A SIMPLE BLOCKADED RYDBERG GAS MODEL

PAIR LOCALIZATION

PAIR LOCALIZATION

The Case of No Blockade

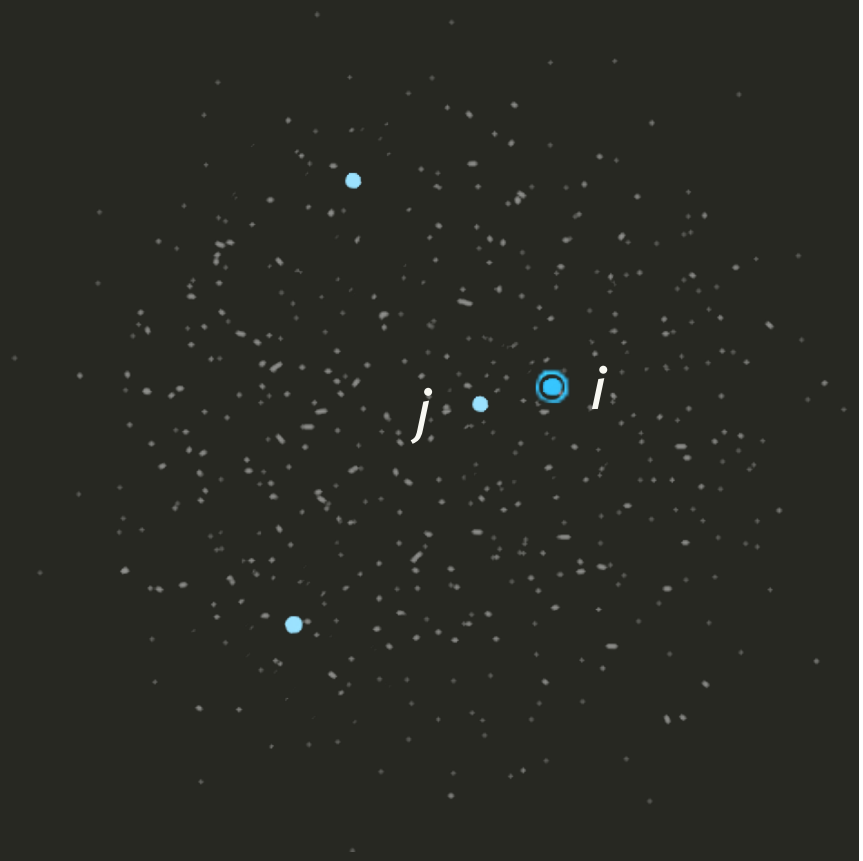
- atoms i and j separated by much less than $(2\pi\rho)^{-1/3}$
- huge coupling strength
- pair-localized eigenstates:

$$\frac{1}{\sqrt{2}}\left(|i\rangle \pm |j\rangle\right)$$

with eigenvalues

$$\Lambda_{\pm} \simeq \pm H_{ij},$$

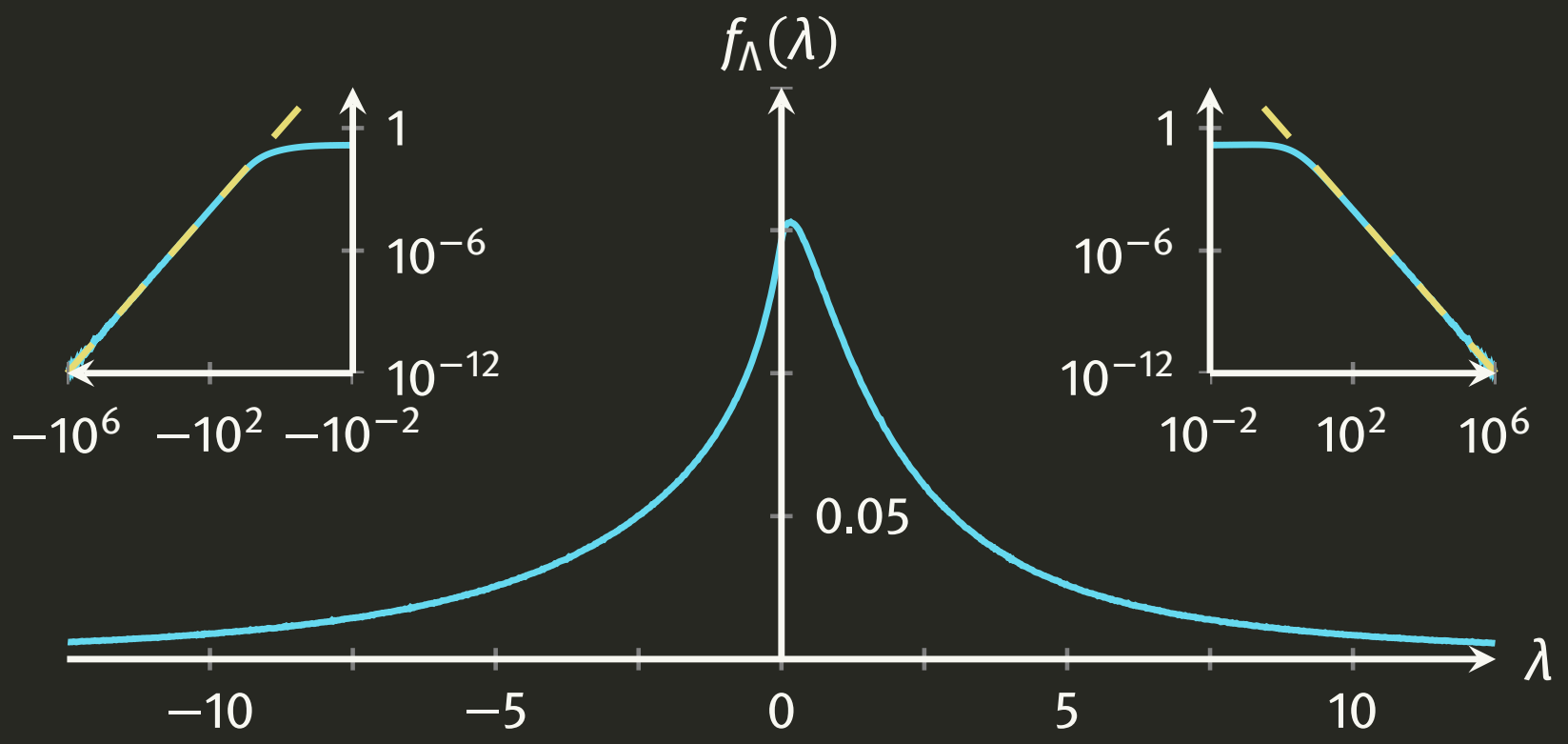
- although rare it is statistically important



PAIR LOCALIZATION

The Case of No Blockade

- density of states: $f_\Lambda(\lambda) = \frac{1}{N}\overline{\mathrm{Tr}[\delta(\lambda - H)]}$,
  $N = 10^4$, $r_\mathrm{b} = 0$

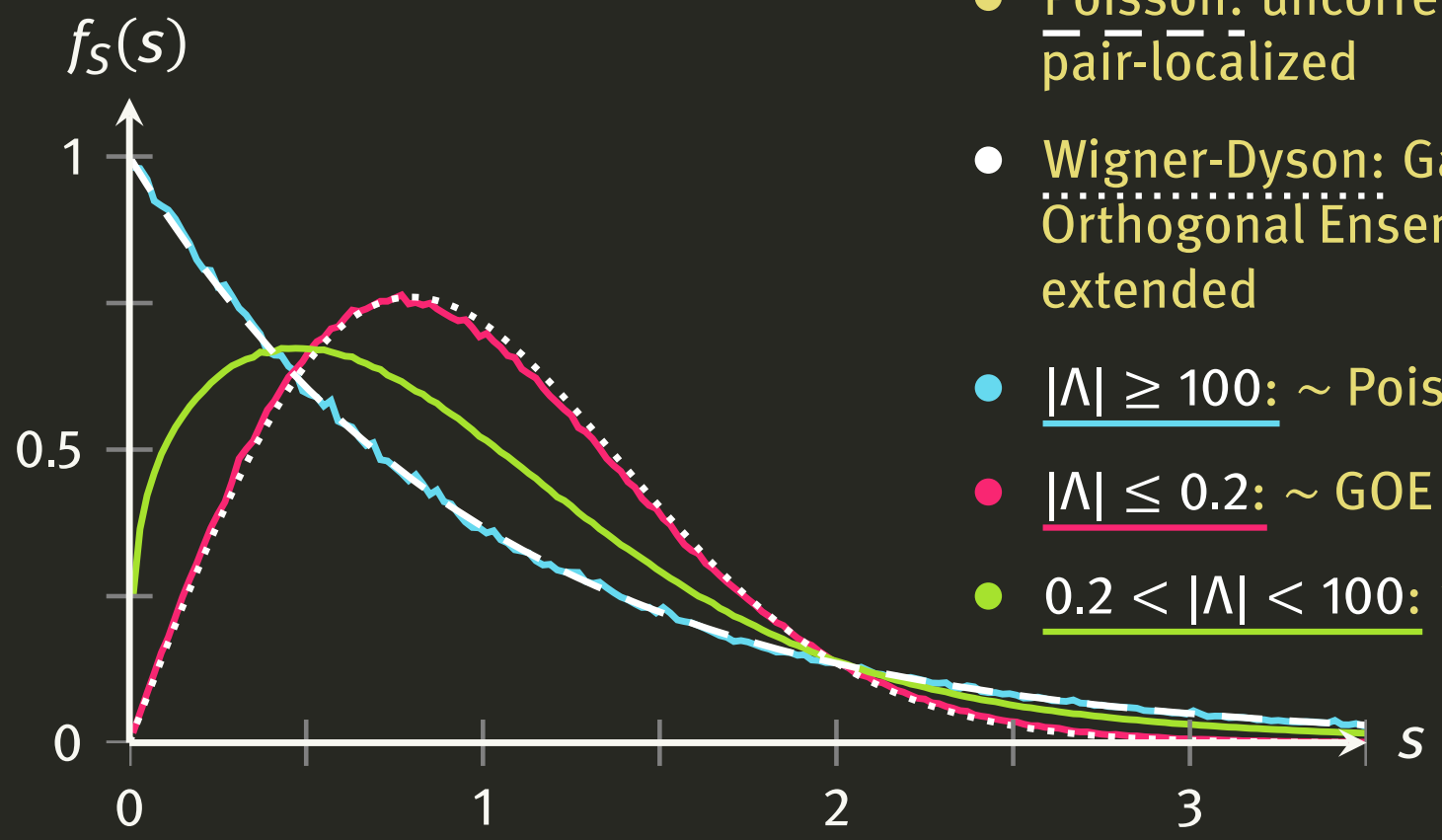


- heavy-tailed: $f_\Lambda(\lambda) \sim_{|\lambda|\to\infty} |\lambda|^{-2}$

PAIR LOCALIZATION

The Case of No Blockade

- level spacing density $f_S(s)\,\mathrm{d}s$: probability to sample two adjacent eigenvalues Λ_ν, $\Lambda_{\nu+1}$ separated by $S \in (s, s+\mathrm{d}s)$, $N = 10^4$, $r_\mathrm{b} = 0$

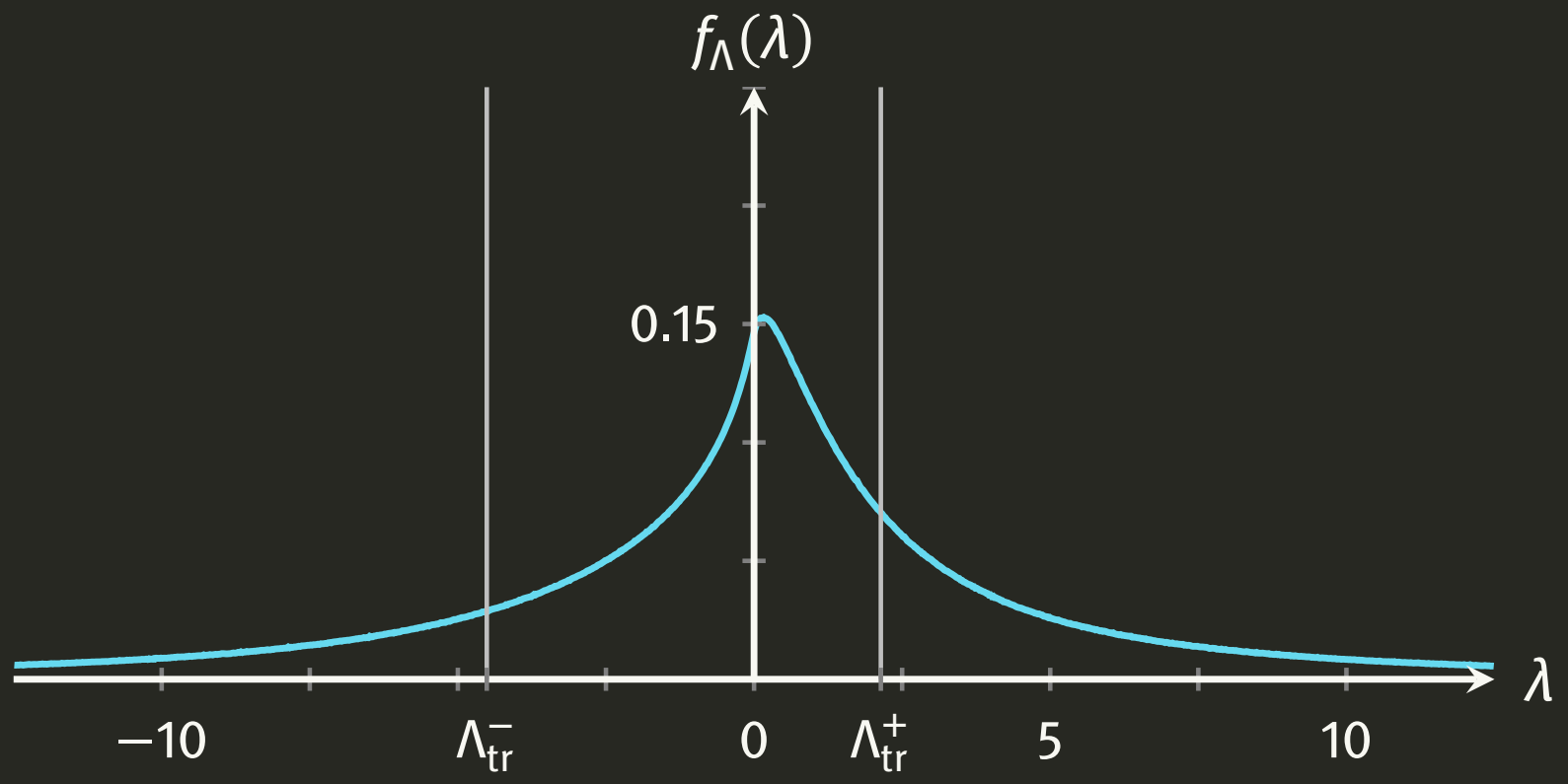


- Poisson: uncorrelated, pair-localized
- Wigner-Dyson: Gaussian Orthogonal Ensemble, extended
- $|\Lambda| \geq 100$: ~ Poisson
- $|\Lambda| \leq 0.2$: ~ GOE
- $0.2 < |\Lambda| < 100$: mixed

PAIR LOCALIZATION

The Case of No Blockade

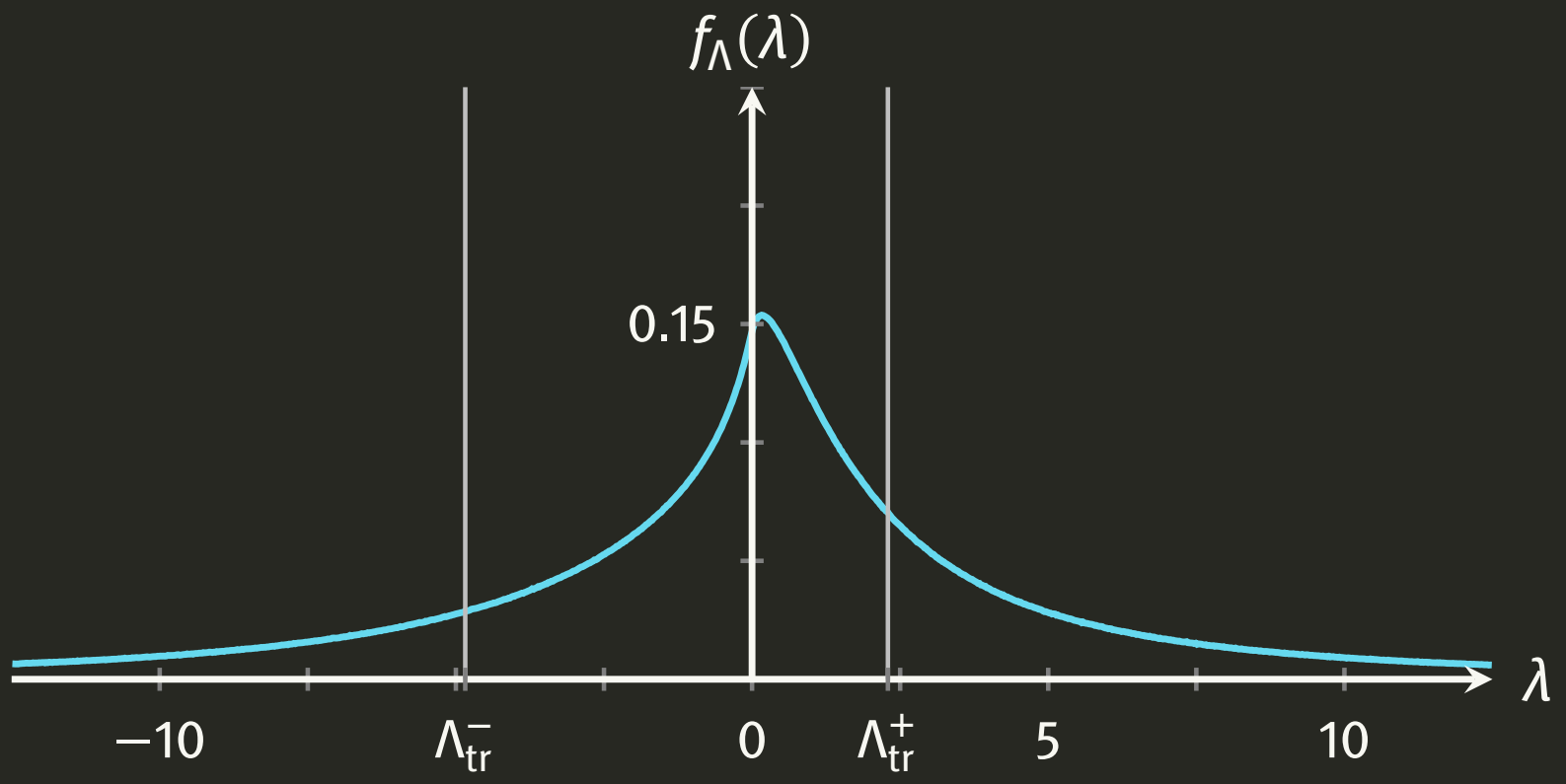
- transition energies: $\Lambda_{\mathrm{tr}}^{\pm}$,
  $N = 10^4$, $r_{\mathrm{b}} = 0$



PAIR LOCALIZATION

The Case of Weak Blockade

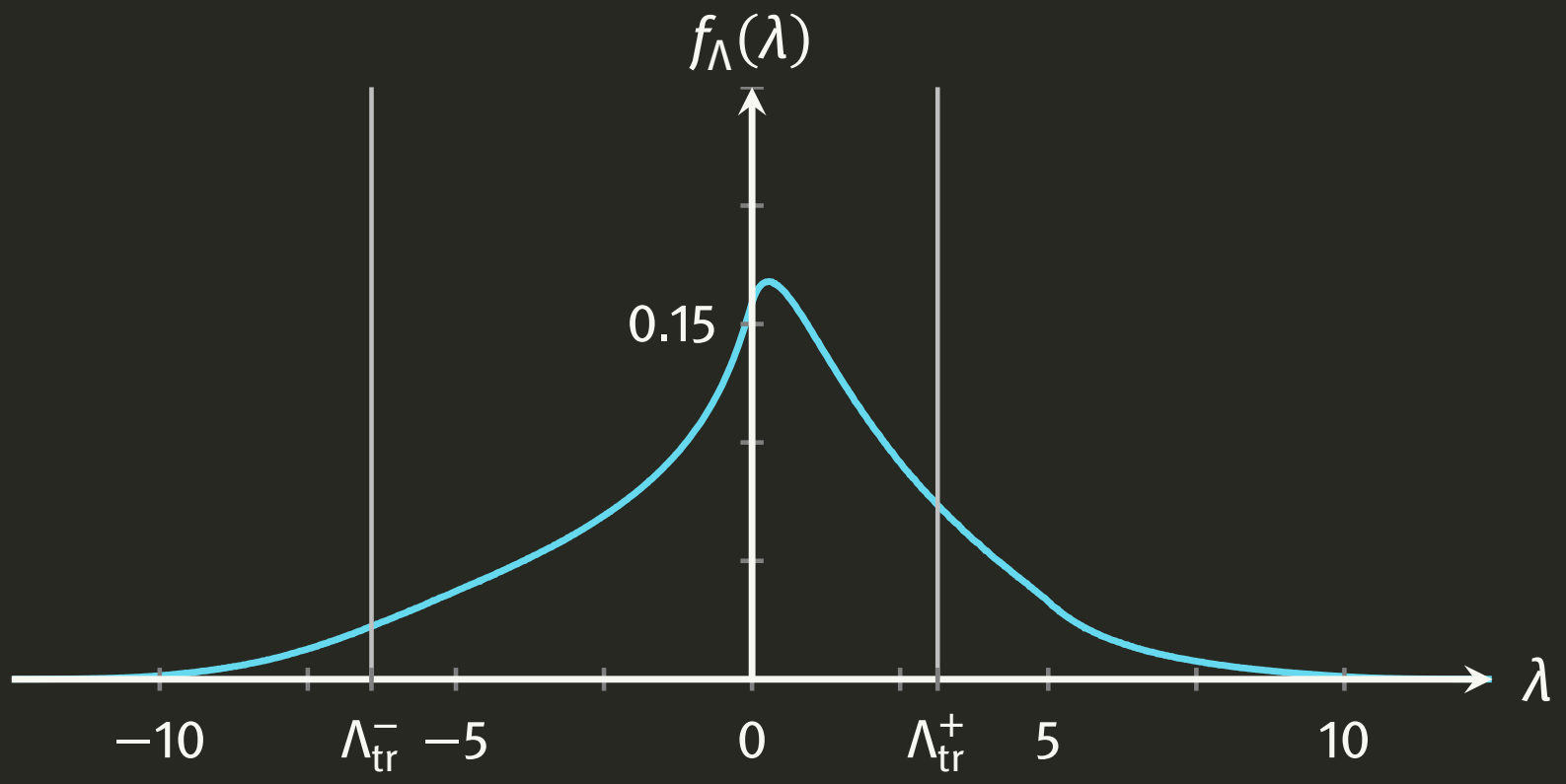
- transition energies: $\Lambda_{\mathrm{tr}}^{\pm}$,
  $N = 10^4$, $r_{\mathrm{b}} = 0.25$



PAIR LOCALIZATION

The Case of Semi-Strong Blockade

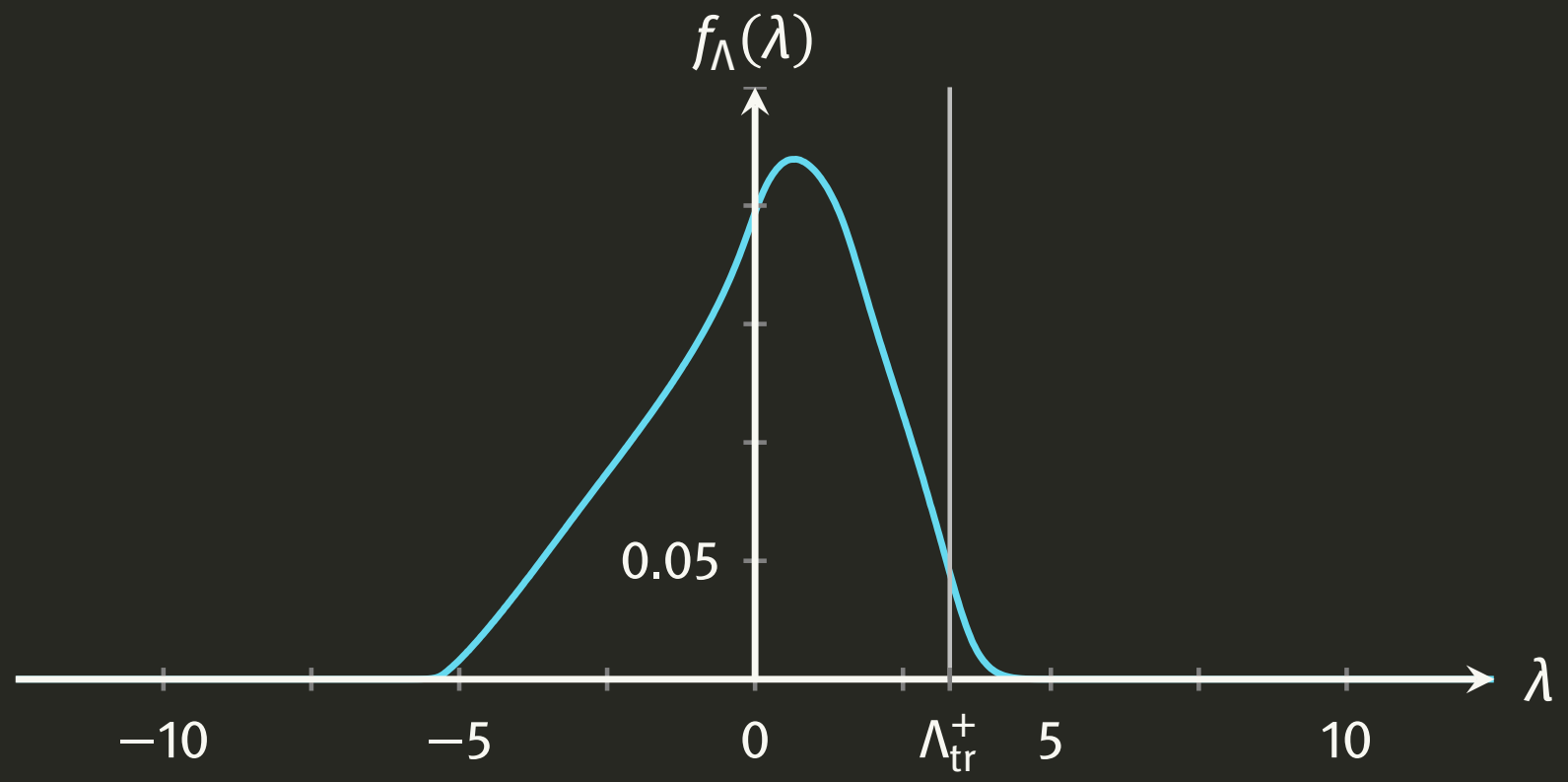
- transition energies: $\Lambda_{\mathrm{tr}}^{\pm}$,
  $N = 10^4$, $r_{\mathrm{b}} = 0.5 \simeq (2\pi\rho)^{-1/3}$



PAIR LOCALIZATION

The Case of Strong Blockade

- transition energies: $\Lambda_{\mathrm{tr}}^{\pm}$,
  $N = 10^4$, $r_{\mathrm{b}} = 0.75 > (2\pi\rho)^{-1/3}$



$\Rightarrow$ dipole blockade controls localization effects!

CONCLUSION

- pair localization is key in energy transfer in Rydberg gases
- transitions from delocalized to localized states

- more blockade → more order & larger minimal distances → less localization → more diffusion
- less blockade → less order & smaller minimal distances → more localization → less diffusion

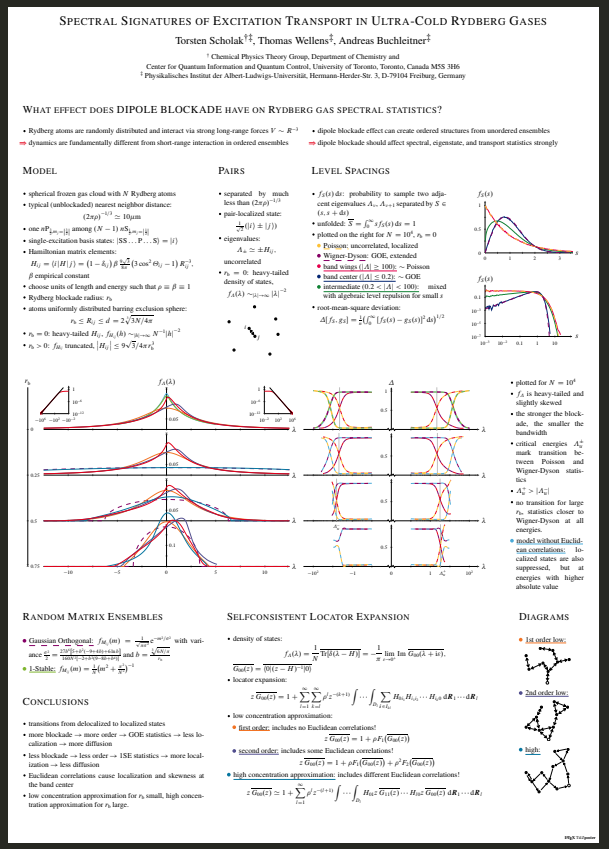
⇒ dipole blockade controls localization effects!

That's all Folks!

POSTER SESSION

On Thursday

- effect of Euclidean correlations in the matrix elements H_{ij}
- analytical treatments for the asymptotical limit $N \to \infty$
  - self-consistent Matsubara-Toyozawa locator expansion for the resolvent $G(z)$
  - a-stable random matrix theory



SPECTRAL SIGNATURES OF EXCITATION TRANSPORT IN ULTRA-COLD RYDBERG GASES

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WHAT EFFECT DOES DIPOLE BLOCKADE HAVE ON RYDBERG GAS SPECTRAL STATISTICS?

- Rydberg atoms are randomly distributed and interact via strong long-range forces $V \sim R^{-3}$
  ⇒ dynamics are fundamentally different from short-range interaction in ordered ensembles
- dipole blockade effect can create ordered structures from unordered ensembles
  ⇒ dipole blockade should affect spectral, eigenstate, and transport statistics strongly

MODEL

- spherical frozen gas cloud with N Rydberg atoms
- typical (unblockaded) nearest neighbor distance: $(2\pi\rho)^{-5/3} \simeq 10\mu\text{m}$
- one $n\mathbf{P}_{|m_\ell=|\mathbf{i}|}$ among $(N-1)$ $n\mathbf{S}_{|m_\ell=|\mathbf{i}|}$
- single-excitation basis states: $|\mathbf{S}S\ldots\mathbf{P}\ldots\mathbf{S}\rangle = |i\rangle$
- Hamiltonian matrix elements: $H_{ij} = \langle i|H|j\rangle = (1-\delta_{ij})\,\beta\,\frac{\|\mathbf{R}_{ij}\|^3}{\mathbf{R}_{ij}^2}\left(3\cos^2\Theta_{ij}-1\right)\mathbf{R}_{ij}^{-3}$, β empirical constant
- choose units of length and energy such that $\rho \cong \beta \cong 1$
- Rydberg blockade radius: r_b
- atoms uniformly distributed barring exclusion sphere: $r_b \le R_{ij} \le d = 2\sqrt[3]{3N/4\pi}$
- $r_b = 0$: heavy-tailed H_{ij}, $f_{H_{ij}}(h) \sim_{|h|\to\infty} N^{-1}|h|^{-2}$
- $r_b > 0$: $f_{H_{ij}}$ truncated, $|H_{ij}| \le 9\sqrt{3}/4\pi r_b^3$

PAIRS

- separated by much less than $(2\pi\rho)^{-1/3}$
- pair-localized state: $\frac{1}{\sqrt{2}}(|i\rangle \pm |j\rangle)$
- eigenvalues: $\Lambda_\pm \simeq \pm H_{ij}$, uncorrelated
- $r_b = 0$: heavy-tailed density of states, $f_\Lambda(\lambda) \sim_{|\lambda|\to\infty} |\lambda|^{-2}$

LEVEL SPACINGS

- $f_S(s)\,\mathrm{d}s$: probability to sample two adjacent eigenvalues Λ_n, Λ_{n+1} separated by $S \in (s, s+\mathrm{d}s)$
- unfolded: $\overline{S} = \int_0^\infty s\, f_S(s)\,\mathrm{d}s = 1$
- plotted on the right for $N = 10^4$, $r_b = 0$
  - Poisson: uncorrelated, localized
  - Wigner-Dyson: GOE, extended
  - band wings ($|\Lambda| \ge 100$): ~ Poisson
  - band center ($|\Lambda| \le 0.2$): ~ GOE
  - intermediate ($0.2 < |\Lambda| < 100$): mixed with algebraic level repulsion for small s
- root-mean-square deviation: $\Delta[f_S, g_S] = \frac{1}{2}\left(\int_0^\infty |f_S(s) - g_S(s)|^2\,\mathrm{d}s\right)^{1/2}$

- plotted for $N = 10^4$
- f_Λ is heavy-tailed and slightly skewed
- the stronger the blockade, the smaller the bandwidth
- critical energies $\Lambda_n^\pm$ mark transition between Poisson and Wigner-Dyson statistics
- $\Lambda_n^+ > |\Lambda_n^-|$
- no transition for large r_b, statistics closer to Wigner-Dyson at all energies.
- model without Euclidean correlations: localized states are also suppressed, but at energies with higher absolute value

RANDOM MATRIX ENSEMBLES

- Gaussian Orthogonal: $f_{H_{ij}}(m) = \frac{1}{\sqrt{2\pi}\sigma}\mathrm{e}^{-m^2/2\sigma^2}$ with variance $\frac{\sigma^2}{N} = \frac{2N^2[1+\delta^2(1-9\alpha+6\alpha\gamma)]}{\pi\sin(\gamma-2\alpha)(N-4\alpha+\alpha\gamma)}$ and $b = \frac{\gamma\sqrt{N/2}}{r_b}$
- 1-Stable: $f_{H_{ij}}(m) = \frac{1}{\pi}\left(m^2 + \frac{b^2}{N^2}\right)^{-1}$

CONCLUSIONS

- transitions from delocalized to localized states
- more blockade → more order → GOE statistics → less localization → more diffusion
- less blockade → less order → ISE statistics → more localization → less diffusion
- Euclidean correlations cause localization and skewness at the band center
- low concentration approximation for r_b small, high concentration approximation for r_b large.

SELFCONSISTENT LOCATOR EXPANSION

- density of states: $f_\Lambda(\lambda) = \frac{1}{N}\overline{\mathrm{Tr}[\delta(\lambda - H)]} = -\frac{1}{\pi}\lim_{\varepsilon\to 0^+}\mathrm{Im}\,\overline{G_{00}(\lambda+\mathrm{i}\varepsilon)}$, $\overline{G_{00}(z)} = \overline{[0|(z-H)^{-1}|0]}$
- locator expansion: $z\,\overline{G_{00}(z)} = 1 + \sum_{\ell=1}^{\infty}\sum_{k=1}^{\infty}\rho^\ell z^{-(k+1)}\int\cdots\int_{B_\ell}\sum_{i\neq i_\ell}\overline{H_{0i_1}H_{i_1i_2}\cdots H_{i_\ell 0}}\,\mathrm{d}\mathbf{R}_1\cdots\mathrm{d}\mathbf{R}_\ell$
- low concentration approximation:
  - first order: includes no Euclidean correlations! $z\,\overline{G_{00}(z)} = 1 + \rho F_1(\overline{G_{00}(z)})$
  - second order: includes some Euclidean correlations! $z\,\overline{G_{00}(z)} = 1 + \rho F_1(\overline{G_{00}(z)}) + \rho^2 F_2(\overline{G_{00}(z)})$
  - high concentration approximation: includes different Euclidean correlations! $z\,\overline{G_{00}(z)} \simeq 1 + \sum_{\ell=1}^{\infty}\rho^\ell z^{-(\ell+1)}\int\cdots\int_{B_\ell}\overline{H_{0i_1}z\,\overline{G_{11}(z)}\cdots H_{i_\ell 0}z\,\overline{G_{00}(z)}}\,\mathrm{d}\mathbf{R}_1\cdots\mathrm{d}\mathbf{R}_\ell$

DIAGRAMS

- 1st order low:
- 2nd order low:
- high:

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